

**Reinforcing the Untrainable: An Applied Behavioral Analysis of Domestic Cat Training
Using Positive Reinforcement Methods**

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Abstract

Given the stereotype that cats are untrainable, this study applied principles of Applied Behavior Analysis (ABA) to test whether cats can learn complex behaviors through positive reinforcement. Training grounded in ABA has long improved animal welfare and strengthened human–animal bonds. This study examined whether an ABA intervention could increase the frequency of trained behaviors in response to commands. The subject was a 14-year-old spayed domestic longhair cat trained by her owner using a single-subject AB design that compared baseline and intervention phases across daily 5–15 minute sessions. Reinforcement included clicker and treat pairing. Target behaviors were *stay*, *back-up*, and *scratch*, and the trainer recorded the percentage of correct responses per cue. Results indicated that *stay* and *back-up* behaviors reached mastery, whereas *scratch* did not. Findings suggest that cats can learn complex tasks through ABA methods, though success may vary depending on motivation, behavior type, and temperament. Overall, the study demonstrates that positive reinforcement can improve feline learning, welfare, and adaptability while challenging stereotypes about cats’ independence and trainability.

Introduction

Picture this: someone says “sit,” and instantly a dog comes to mind. But why not a cat? Despite sharing our homes and hearts for thousands of years, cats are rarely trained or treated like dogs. That observation led me to wonder how much of this difference stems from the animals themselves and how much comes from our expectations of them. What causes so many people to train dogs versus cats and is the reasoning supported by evidence?

There are many reasons for this difference in training capabilities, but one possible explanation is that cats are behaviorally closer to their wildcat ancestry compared to other domesticated species such as dogs. Genetic research supports this idea, showing that domestic cats remain closely related to their wild counterparts and share many behavioral traits of wildcats, reflecting a comparatively incomplete domestication process (Lyons, 2012). This idea that cats are closer to their wildcat roots leads to the belief that they are more independent from their human owners and supports stereotypes that cats are “untrainable.” However, a simple fact disputes this theory: owners are able to recognize and understand their cat’s meows, which demonstrates a strong capacity for learning and communication from both sides (Calderone, 2015). The Lexington Attachment to Pets Scale (LAPS) demonstrated that dog owners had higher scores in every country, and previous studies have shown that cat owners displayed less emotional attachment and were less willing to pay for services that benefited their cats compared to dog owners, which many researchers attribute to behavioral differences (Sandøe, 2023). Conversations with veterinarians have also shown that people report feline behavior problems more often than canine ones, but they are less likely to manage those problems compared to dog owners (Hevern, 2022). This likely reinforces the misconception that cats are not trainable, since under-investing in solutions will naturally lead to a lack of progress, especially in comparison to the training programs in place for dogs. These long-standing perceptions about cats’ independent nature have also influenced which animals were historically bred and trained for specific purposes.

Looking at past work and real-world examples further highlights this divide. While dogs have long histories of training and service (Overall & Dyer, 2005), cats do not. Due to the specific skills and tasks required of service dogs, cats are often not chosen; this is primarily

because of the history of selective breeding for service in dogs rather than a true lack of trainability in cats. In fact, cats have already proven themselves to be domesticated and adaptable across environments, which indicates that they have a strong capacity to learn new behaviors despite persistent stereotypes (Lyons et al., 2012). Together these findings suggest that social perception, not biology, limits cat training.

Building on that idea, empirical research shows that cats can, in fact, learn specific behaviors through structured, reward-based methods. Clicker training and environmental enrichment are particularly effective, whereas forceful approaches tend to fail (Overall & Dyer, 2005). Applied Behavior Analysis (ABA)—the systematic use of reinforcement to shape voluntary behavior—underlies many of these successful techniques. For example, shelter cats have been trained to perform specific tasks and tricks using positive reinforcement (Willson et al., 2017), and even horses trained with clickers demonstrate stronger resistance to extinction, suggesting that ABA principles generalize across species (Williams et al., 2004). Likewise, zoo animals traditionally considered difficult to handle respond reliably to operant-conditioning programs, further reinforcing that cats are not an exception. Studies using accelerometer data have also been able to classify distinct feline behaviors, showing that learned behaviors can be measured, tracked, and refined objectively (Watanabe et al., 2005). Together, this evidence demonstrates that cats are not only capable of learning complex behaviors but that the same learning principles that apply to other species apply to them as well. In other words, the science confirms what experience often demonstrates; cats are not resistant to learning, but they do require patience and positive methods

While this evidence confirms cats' ability to learn, it is not enough to simply show that cats can be trained; it is equally important to understand why training is beneficial to their overall

well-being. Training methods allow cats to practice natural hunting and play behaviors, improving engagement, exercise, and welfare (Overall & Dyer, 2005). Indoor cats often have less access to natural engagement, which can lead to stress (Jayne & See, 2019). Enrichment combats the stress caused by sterile or restricted environments, particularly in zoo settings, training has been shown to increase environmental enrichment, which in turn reduces negative or stereotypic behaviors (Fernandez & Martin, 2021).

Additionally, cats are a more independent species of domesticated pets and, thus, they may develop illnesses or behavioral issues when they feel they lack control, which training solves by providing stimulation and a sense of control (Buffington & Bain, 2020). Welfare also includes temperament considerations. Cats with aggressive or dominant tendencies benefit from structured enrichment, which decreases problem behaviors and builds confidence (Travnik et al., 2020; Rubenstein, 2021; Fernandez & Martin, 2021). Structured routines, such as daily scheduled training, help cats adjust and adapt to as can be seen with feral cats in managed colonies adapting to feeding schedules by humans (Hernandez et al., 2018) and shelter dogs improving welfare under daily structure (Tuber et al., 1999). Predictable routines increase both activity levels and the bond between trainer and animal (Fernandez & Martin, 2021). This indicates the importance of providing structure and enrichment activities, including training, for cats as the welfare benefits are clear.

However, beyond simply participating in training, procedure and methods are extremely important to generate persistent skills that are applicable to a variety of contexts. Training methods have shifted from coercive techniques to positive reinforcement. In this approach, a reinforcer is provided immediately after a correct response, increasing the likelihood the behavior will recur (Rubenstein, 2021). Modern zoo training is rooted in operant conditioning,

which is a learning process where voluntary behaviors are followed by a reinforcement or punishment with the goal of changing the likelihood the behavior will occur again and is directly related to concepts in behaviorism. In shelter dogs, conditioning paired with human contact reduced stress (Tuber et al., 1999). Beyond welfare and behavioral considerations, neuroscience also supports reinforcement as a universal learning mechanism. From this perspective, dopamine and reward systems drive motivation, validating reinforcement-based training and ABA methods (Newquist & Gardner, 2015). Neuroscience also shows why reinforcement is effective, since brain-behavior pathways can be targeted to address problem behaviors (Amadei et al., 2021).

ABA has also been applied successfully in feline contexts, using enrichment and operant analysis to modify behavior and resolve issues such as aggression (Overall & Dyer, 2005; Fernandez & Martin, 2021). However, the research on cats using ABA principles is lacking, especially compared to the same research on dogs. There is currently a research gap on directly applying reinforcement techniques to cats which is something that should be rectified.

Training not only produces short-term learning but also long-term emotional and behavioral benefits. Training has been shown to strengthen the bond between cats and their owners by reinforcing emotional connections and communication (Yuhas, 2015). Also, those basic skills transfer to real-world situations, for example, enrichment activities and agility improve adaptability to new stimuli and environments (Overall & Dyer, 2005). Training zoo animals for voluntary husbandry tasks, such as blood draws or exams, also shows that training produces lasting welfare benefits (Fernandez & Martin, 2021). Specifically, in cats, research validating stress measures found that negative handling experiences increase stress in future veterinary visits (Moody et al., 2018). From this, we can infer that positive, gentle handling reduces long-term stress and improves veterinary care experiences. These benefits persist even

after active training decreases. Once cats master a behavior, reinforcement frequency can be reduced, and the behavior will still continue (Rubenstein, 2021). Given cats' long lifespans, structured care, such as training, has long-lasting impacts (Kent et al., 2022). Conversely, dogs are often returned to shelters due to unresolved behavior problems, which shows the importance of persistence in improving behaviors (Tuber et al., 1999). The success and persistence of behaviors also depend on the cat's temperament, highlighting the importance of tailoring training approaches to each specific cat which follows applied behavioral analysis training (Travnik et al., 2020).

Finally, training is not only about welfare but also serves as a valid research approach. Reinforcement is a fundamental psychological principle, central to learning models and human motivation, and cats share the same neurological principles that underlie these behaviors. Homes can act as research labs, with structured routines emulating welfare improvements seen in shelter environments (Tuber et al., 1999). Instead of bringing cats into laboratories, individuals can study them in their natural environment. For example, cameras have been used in TNR colonies to study real-life cat behavior (Hernandez et al., 2018). Principles of behavior analysis were historically tested on animals such as in Skinner's experiments, which later became foundational to psychology as a field (Fernandez & Martin, 2021). Cats themselves have long been used in psychological research, making them valid subjects in humane, enrichment-based studies today (Parrish, 1990) and in helping scientists better understand the brain and cognitions behind behaviors.

Therefore, my study draws on all of the above ideas in addressing the question: Does this ABA-oriented intervention (cat training using positive reinforcement) increase this behavior (cat tricks)? I will be conducting a behavioral analysis experiment on a 14-year-old spayed female cat

in a naturalistic environment. My hope is to demonstrate the overall trainability of cats and counter stereotypes of independence and wild ancestry, improve my cat's well-being by reducing stress and enhancing adaptability, effectively demonstrate principles of ABA and positive reinforcement, and provide long-term benefits so these skills remain with my cat and transfer to other useful behaviors, such as vet visits.

Method

Subject

Kisses, a 14-year-old spayed female domestic longhair cat, served as the subject. She had prior clicker-training experience and lived indoors with the researcher. Full breed is unknown due to unknown genetics and heritage. Kisses had previous experience with basic commands such as sitting, spinning, and responding to her name. To ensure continuous consent—since the subject cannot verbally state “no” but can clearly indicate when she wishes to discontinue—the researcher implemented cooperative care techniques, allowing the cat to leave if stressed and ensuring all participation was voluntary throughout training (Bessant et al., 2022).

Materials

Behavioral data were recorded after each cue using a phone or notepad, then entered into Excel for analysis. All training sessions took place in the researcher's bedroom. The reinforcers used were Temptations brand hard treats and Inaba Churu tubes. Because Kisses is slightly overweight, the researcher limited the number of treats and used a clicker to reinforce correct responses after a variable number of cues. Kisses had already been clicker-trained, which is explained further in the procedure section. Additional materials included a corrugated cardboard scratcher designed for cats. The trainer, who is also the researcher, has extensive experience

training cats and dogs through fostering, volunteering, and professional pet-sitting, providing experience with diverse behaviors and reinforcement situations.

Procedure

Training occurred daily for 5-15 minutes at a consistent time in the same indoor environment on weekdays. The study used an AB design following the Applied Behavior Analysis (ABA) approach. An AB design includes a baseline (A) phase followed by an intervention (B) phase without withdrawal. The baseline period lasted 10 days, and the intervention lasted 24 days. The independent variable was the progression of training days within the intervention phase, and the dependent variable was the percentage of correct behavioral responses. Due to the study design, once the intervention began, the study was not returned to baseline. Phase change occurred after a period of stable responding, assessed and approved by a trained ABA professional. During the baseline phase, the researcher gave the cues but did not provide explicit reinforcement. The researcher waited to see whether Kisses would perform the target behaviors. The baseline phase was also used to strengthen the clicker–treat pairing; the researcher sounded the clicker and immediately gave a treat, associating the clicker with reinforcement through classical conditioning while limiting caloric intake.

During the intervention phase, the researcher stated one of three cues—“scratch,” “back-up,” or “stay”—which served as antecedents for the target behaviors. Each session included all three cues and training in each behavior, presented in a pseudo-random order to avoid sequence effects or shared associations. After the cue “scratch,” Kisses scratched only on the designated scratching board using her nails to produce a scratching sound. After the cue “back-up,” Kisses took at least one full step backward, with each paw moving backward at least one inch. After the cue “stay,” Kisses remained in place in a sitting position for at least three seconds. The

dimension measured for all three behaviors was the frequency of correct behaviors divided by the number of behavior cues given.

All behaviors required that the researcher not physically touch or guide the cat. For example, “back-up” required no physical pushing, “scratch” required no paw manipulation, and “stay” required no physical restraint. During the intervention phase, a reinforcement contingency was used. If Kisses correctly completed a behavior following the antecedent cue, an immediate clicker sound served as the conditioned reinforcer. To maintain a strong clicker–treat association, a variable-ratio reinforcement schedule was used, so Kisses received a treat following the clicker after an average number of correct behaviors. To ensure measurement reliability and consistency, the researcher reviewed data entries immediately after each session to confirm accuracy in recording. Inter-observer validity was also checked periodically by an ABA-trained professional, and, although they could not confirm the training behaviors, they validated that data were being recorded consistently and correctly. The same tone and volume were used for all verbal cues during the approximately five- to fifteen-minute training sessions. For the “stay” behavior, the researcher gradually increased the distance from Kisses across sessions to strengthen stimulus control and generalization. This maintained consistency across sessions and increased the validity of the study.

Two brief data hiatus periods occurred and are indicated in the graph. The first occurred during fall break when the researcher could not consistently train at the same time each day. The second occurred when Kisses developed a serious urinary tract infection requiring antibiotics and gabapentin for pain. To maintain welfare and research consistency, the researcher paused the study for five days until Kisses had recovered and regained strength. To limit bias, the researcher

defined the dimensions and operational definitions of all behaviors before the study began.

Attrition would occur if Kisses became unwilling or unable to continue participating in training.

Results

The purpose of this study was to evaluate whether a positive reinforcement-based ABA intervention increased the frequency of correct behavioral responses to a cue (“scratch”, “back-up”, “stay”) in a domestic cat. Figure 1 displays the percentage of correct responses for each target behavior across baseline and intervention sessions. During the baseline phase, responses remained low and stable across all three behaviors. Following the introduction of the intervention involving treats for correct responses to cues, accuracy continually increased for both the “scratch” and “stay” behaviors. Overall, the data indicated a clear improvement in performance of the specific behaviors across the intervention period.

It is important to note that the days on the x-axis represent study days (weekdays only), with one training session per day. Between days 19–23, the subject was diagnosed with a severe urinary tract infection that required emergency veterinary attention and antibiotics, which caused side effects such as drowsiness. Medication continued until day 29, but the researcher determined her symptoms had lessened enough to resume on day 22 of the study. There was also a brief two-day hiatus prior to the infection when the researcher was away.

Across behaviors, the percentage of correct responses increased from an average of 0% in the baseline stage to 41% in the intervention stage (see Figure 1). Figure 2 presents the “stay” behavior, which increased from an average of 0% correct responses to 70%. Figure 3 shows the “back-up” behavior, which rose from an average of 0% in baseline to 54% correct responses. Although variability increased during the intervention, this likely reflected active acquisition of

the target behaviors. The final graph, Figure 4, depicts the “scratch” behavior. The slope from the baseline to intervention phase remained stable and did not change from a 0% average of accurate responses, keeping the level constant across both phases. Overall, the intervention produced clear increases in two of the three trained behaviors, demonstrating consistent improvement in response accuracy during the intervention phase.

Discussion

Main Findings

This study investigated whether an ABA-oriented intervention (positive reinforcement) could effectively increase specific trained behaviors in a feline. The specific behaviors were stay, back-up, and scratch with a female, spayed, and domesticated 14-year-old cat. The goal of this study was to counter stereotypes of cats’ untrainability and demonstrate that, despite their wild ancestry and beliefs that cats are too independent to learn tricks, owners can improve their cats’ well-being and reduce stress by training their cats using ABA-principles that could hopefully help these behaviors transfer to other useful aspects of their lives. Overall, the study was a success with the cat demonstrating a knowledge of 2 of the 3 cat tricks and was able to consistently complete them in response to the cue. The behavior of stay showed significant improvement with a high amount of accuracy and an ability to generalize to greater and greater distances throughout the training. The behavior of back-up showed moderate to high improvement but was slightly more variable and less stable than stay. The last behavior of scratch was ultimately unsuccessful in training and never fully achieved in response to the trainer’s cue despite multiple alterations of the command. These results partially support the

hypothesis; while two behaviors increased as hypothesized, the scratching behavior did not show measurable improvement under the intervention conditions.

Implications

As demonstrated in previous studies, cats can be effectively clicker-trained to perform different tasks in relatively short periods of time, which is consistent with prior evidence that cats can learn specific behaviors through reinforcement (Willson et al., 2017; Smith et al., 2008). Many studies also indicate that cats with strong existing relationships with their humans have an even greater potential for successful clicker training (Kogan & Schoenfeld-Tacher, 2017). The clear increases in correct behavioral responses in this study reflect the effectiveness of Applied Behavior Analysis (ABA) and demonstrate how it can be generalized for pet training when reinforcers are appropriately selected and timed. However, individual differences in temperament, motivation, and prior experiences may influence how consistently cats respond to reinforcement, highlighting the importance of tailoring training approaches to each cat's behavioral profile (Travnik et al., 2020).

When considering why the specific scratching behavior was unsuccessful, it is important to recognize that scratching is an innate feline behavior used to groom claws and leave both visual and scent markers. Because this behavior is internally motivated and provides its own positive sensory feedback, it is less responsive to external cues. The goal of this study was to increase scratching on designated surfaces to reduce scratching on restricted areas, but additional deterrent or redirection strategies may have strengthened this approach. For example, adding sticky tape to restricted surfaces or placing scratching posts near preferred resting areas could have redirected the behavior more effectively (Calderone, 2015; Grigg, 2019); Overall et al., 2005). Scratching, in particular, is a self-rewarding behavior, meaning that treats from the

researcher serve more as a bonus than as the main motivation for the action (Overall et al., 2005). This is especially relevant considering that individual cats differ in their sensitivity to reinforcers, and self-reinforcing behaviors often compete with external rewards (Kogan et al., 2017).

Interestingly, over time the cat began to look for praise and treats after scratching, even when the cue had not been given. This applied not only to the trainer but also to other individuals in the environment, suggesting that the cat learned to associate scratching with positive human attention. Although the command itself was never mastered, the pattern indicates that social reinforcement and improved communication between the cat and trainer developed over the course of the study. This aligns with prior research showing that reinforcement-based interactions both shape behavior and strengthen the emotional bond between cats and humans (Yuhás, 2015).

Strengths & Limitations

This study took place in a naturalistic home environment, which provided high external and ecological validity (Hernández et al., 2018). Because the research setting mirrored a typical owner–pet interaction, the findings have strong practical relevance and replicability for other cat owners interested in clicker training. The procedures were clearly defined and followed specific Applied Behavior Analysis (ABA) operational principles, increasing reliability and transparency. Cooperative care methods were also used to ensure the cat’s assent was maintained throughout the process, as participation was entirely voluntary (Bessant et al., 2022). This approach models how behavioral training can be conducted humanely and effectively while respecting animal welfare, offering a framework for other owners seeking to improve their cats’ behavior through positive reinforcement.

However, as with many ABA studies, the design focused on a single subject, which limits the generalizability of the findings to other cats. While this single-case format allows for detailed functional analysis of behavior, it also increases the risk that individual differences or environmental factors may have influenced results. Another limitation was the limited number of inter- or intra-observer reliability checks, which could have strengthened the objectivity of data collection and reduced potential researcher bias (Travnik et al., 2020). Additionally, the study period was relatively short, restricting the opportunity to implement alternative intervention methods, particularly for the scratching behavior, which might have provided greater insight into reinforcement variation and extinction patterns over time.

Despite these limitations, the study demonstrates that cats can successfully learn trained behaviors using ABA-based positive reinforcement. These findings contribute to countering the widespread stereotype that cats are untrainable and support an evidence-based understanding of feline cognition and welfare. They also have clear practical applications for cooperative care and enrichment in both homes and shelters. Reinforcement-based approaches have been linked to stronger human–animal bonds (Yuhas, 2015) and reduced stress during veterinary care or handling procedures (Moody et al., 2018), suggesting that small-scale interventions like this one can have long-term welfare benefits when applied more broadly.

Future research

Future research could explore a range of ways to apply ABA principles with cats, but one helpful idea would be introducing multiple cats to the same training program to examine individual differences and increase generalizability. It may also be interesting to compare training with and without the use of a clicker or using other reinforcement methods such as food, play, or praise. Temperament-based training approaches could also be explored, as differences in sociability,

boldness, or fearfulness may predict responsiveness to reinforcement and rate of learning (Travnik et al., 2020). Another interesting idea is a side-by-side comparison of cat and dog learning behaviors and the way they learn along with the speed, generalizability, and extinction of behaviors over time. Finally, examining overall welfare indicators of cats who have undergone training versus those who have not could help determine whether positive reinforcement produces measurable improvements in stress, human–animal bonding, and long-term health. This study can contribute to the growing knowledge base that recognizes that, while cats are complex species, they are domesticated and not, in fact, untrainable—just in need of approaches that reflect their unique motivations. Recognizing this within the pet world can reaffirm owner–cat bonds, improve welfare outcomes, and deepen our understanding of animal behavior across species.

References

- Amadei, E., Cantile, C., Gazzano, A., Pierantoni, L., & Mariti, C. (2021). The link between neurology and behavior in veterinary medicine: A review. *Journal of Veterinary Behavior*, 46, 40-53. <https://doi.org/10.1016/j.jveb.2021.07.003>
- Bessant, C., Dowgray, N., Ellis, S. L. H., Rodan, I., & Sparkes, A. H. (2022). ISFM's Cat Friendly Principles for Veterinary Professionals. *Journal of Feline Medicine and Surgery*, 24(11), 1087–1092. <https://doi.org/10.1177/1098612X221128750>
- Buffington, C. A. T., & Bain, M. (2020). Stress and feline health. *Veterinary Clinics of North America: Small Animal Practice*, 50(4), 653–662.
<https://doi.org/10.1016/j.cvsm.2020.03.001>
- Calderone, J. (2015). Decoding the Cat. *Scientific American Mind*, 26(3), 43–43.
<https://www.jstor.org/stable/24946450>
- Fernandez, E. J., & Martin, A. L. (2021). Animal Training, Environmental Enrichment, and Animal Welfare: A History of Behavior Analysis in Zoos. *Journal of Zoological and Botanical Gardens*, 2(4), 531-543. <https://doi.org/10.3390/jzbg2040038>
- Grigg, E. K., & Kogan, L. R. (2019). Owners' Attitudes, Knowledge, and Care Practices: Exploring the Implications for Domestic Cat Behavior and Welfare in the Home. *Animals : an open access journal from MDPI*, 9(11), 978. <https://doi.org/10.3390/ani9110978>
- Hernandez, S. M., Loyd, K. A. T., Newton, A. N., Gallagher, M. 'C.', Carswell, B. L., & Abernathy, K. J. (2018). Activity patterns and interspecific interactions of free-roaming, domestic cats in managed Trap-Neuter-Return colonies. *Applied Animal Behaviour Science*, 202, 63–68. <https://doi.org/10.1016/j.applanim.2018.01.014>

Hevern, M. E. (2022). Welfare worries: A preliminary, cross-sectional study of general practice, small animal veterinarians perceptions and management of canine and feline behaviour problems. *Australian Veterinary Journal*, 100(8), 377-387.

<https://doi.org/10.1111/avj.13168>

Jayne, K., & See, A. (2019). Behavioral Research on Captive Animals: Scientific and Ethical Concerns. In K. Jayne & K. Herrmann (Eds.), *Animal Experimentation: Working Towards a Paradigm Change* (Vol. 22, pp. 517–548). Brill.

<http://www.jstor.org/stable/10.1163/j.ctvjhzq0f.28>

Kogan, L., Kolus, C., & Schoenfeld-Tacher, R. (2017). Assessment of Clicker Training for Shelter Cats. *Animals : an open access journal from MDPI*, 7(10), 73.

<https://doi.org/10.3390/ani7100073>

Lyons, L. A., & Kurushima, J. D. (2012). A short natural history of the cat and its relationship with humans. In S. E. Little (Ed.), *The cat: Clinical medicine and management* (pp. 1254–1262). WB Saunders.

Moody, C. M., Picketts, V. A., Mason, G. J., Dewey, C. E., & Niel, L. (2018). Can you handle it? Validating negative responses to restraint in cats. *Applied Animal Behaviour Science*, 204,

94–100. <https://doi.org/10.1016/j.applanim.2018.04.012>

Newquist, G., & Gardner, R. A. (2015). Reconsidering Food Reward, Brain Stimulation, and Dopamine: Incentives Act Forward. *The American Journal of Psychology*, 128(4), 431–444. <https://doi.org/10.5406/amerjpsyc.128.4.0431>

Overall, K. L., Rodan, I., Beaver, B. V., Carney, H., Crowell-Davis, S., Hird, N., Kudrak, S., & Wexler-Mitchell, E. (2005). Feline behavior guidelines from the American Association of

Feline Practitioners Panel on Feline Behavior Guidelines. *Journal of the American Veterinary Medical Association*, 227(1), 70–84.

<https://doi.org/10.2460/javma.2005.227.70>

Parrish, C. R. (1990). Emergence, natural history, and variation of canine, mink, and feline parvoviruses. In K. Maramorosch, F. A. Murphy, & A. J. Shatkin (Eds.), *Advances in virus research* (Vol. 38, pp. 403–450). Academic Press. [https://doi.org/10.1016/S0065-3527\(08\)60867-2](https://doi.org/10.1016/S0065-3527(08)60867-2)

Rubenstein, E. (2021). Who’s a Good Director? Dog-Training Strategies for Better Rehearsing. *The Choral Journal*, 62(2), 20–29. <https://www.jstor.org/stable/27089730>

Sandøe, P., Palmer, C., Corr, S. A., Springer, S., & Lund, T. B. (2023). Do people really care less about their cats than about their dogs? A comparative study in three European countries. *Frontiers in Veterinary Science*, 10, Article 1237547. <https://doi.org/10.3389/fvets.2023.1237547>

Smith, S. M., & Davis, E. S. (2008). Clicker increases resistance to extinction but does not decrease training time of a simple operant task in domestic dogs (*Canis familiaris*). *Applied Animal Behaviour Science*, 110(3–4), 318–329. <https://doi.org/10.1016/j.applanim.2007.04.012>

Travnik, I. d. C., Machado, D. d. S., Gonçalves, L. d. S., Ceballos, M. C., & Sant’Anna, A. C. (2020). Temperament in Domestic Cats: A Review of Proximate Mechanisms, Methods of Assessment, Its Effects on Human—Cat Relationships, and One Welfare. *Animals*, 10(9), 1516. <https://doi.org/10.3390/ani10091516>

- Tuber, D. S., Miller, D. D., Caris, K. A., Halter, R., Linden, F., & Hennessy, M. B. (1999). Dogs in Animal Shelters: Problems, Suggestions, and Needed Expertise. *Psychological Science*, 10(5), 379–386. <http://www.jstor.org/stable/40063451>
- Watanabe, S., Izawa, M., Kato, A., Ropert-Coudert, Y., & Naito, Y. (2005). A new technique for monitoring the detailed behaviour of terrestrial animals: A case study with the domestic cat. *Applied Animal Behaviour Science*, 94(1–2), 117–131.
<https://doi.org/10.1016/j.applanim.2005.01.010>
- Williams, J. L., Friend, T. H., Nevill, C. H., & Archer, G. (2004). The efficacy of a secondary reinforcer (clicker) during acquisition and extinction of an operant task in horses. *Applied Animal Behaviour Science*, 88(3–4), 331–341.
<https://doi.org/10.1016/j.applanim.2004.03.008>
- Willson, E. K., Stratton, R. B., Bolwell, C. F., & Stafford, K. J. (2017). Comparison of positive reinforcement training in cats: A pilot study. *Journal of Veterinary Behavior*, 21, 64–70.
<https://doi.org/10.1016/j.jveb.2017.07.007>
- Yuhas, D. (2015). PETS: WHY DO WE HAVE THEM? *Scientific American Mind*, 26(3), 28–33.
<https://www.jstor.org/stable/24946447>

Tables, Figures, Appendices

Figure 1

Percentage of Correct Responses for all Behaviors Across Baseline (BL) and Intervention (IV)

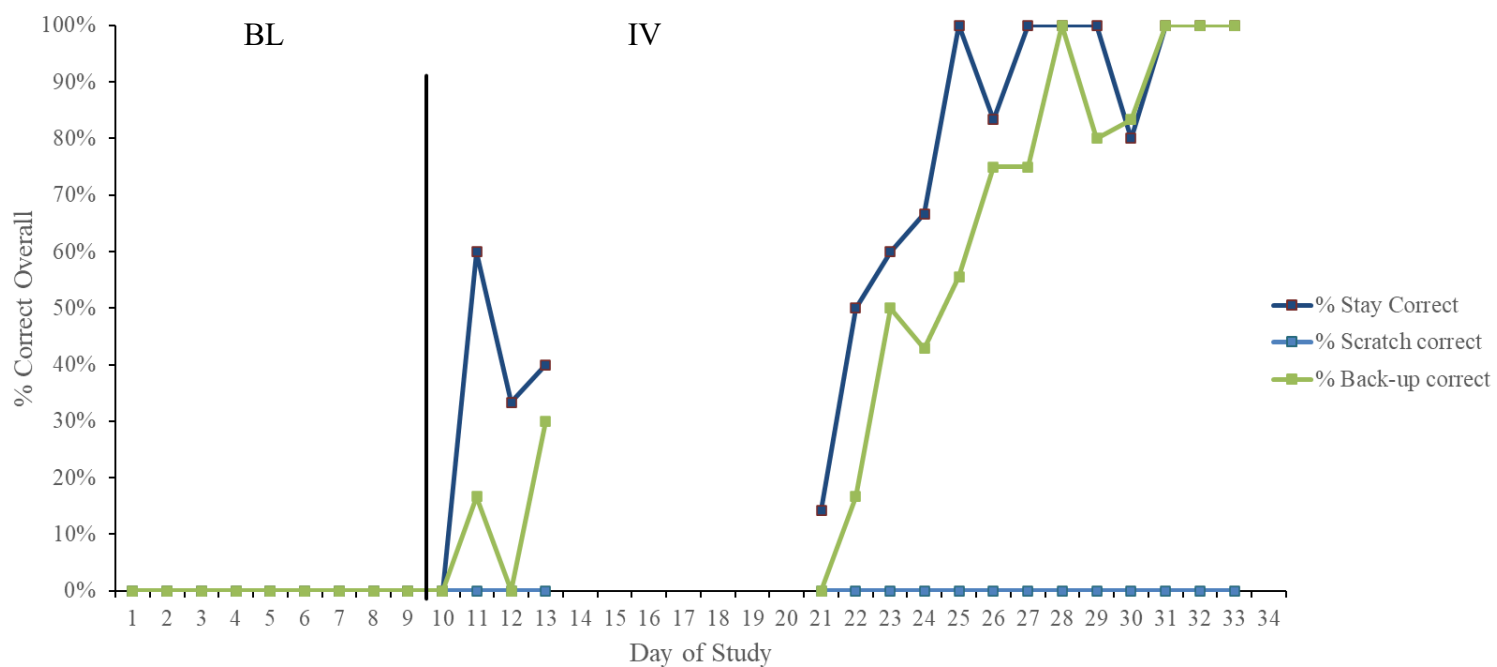


Figure 2

Percentage of Correct Responses for Stay Behavior Across Baseline (BL) and Intervention (IV)

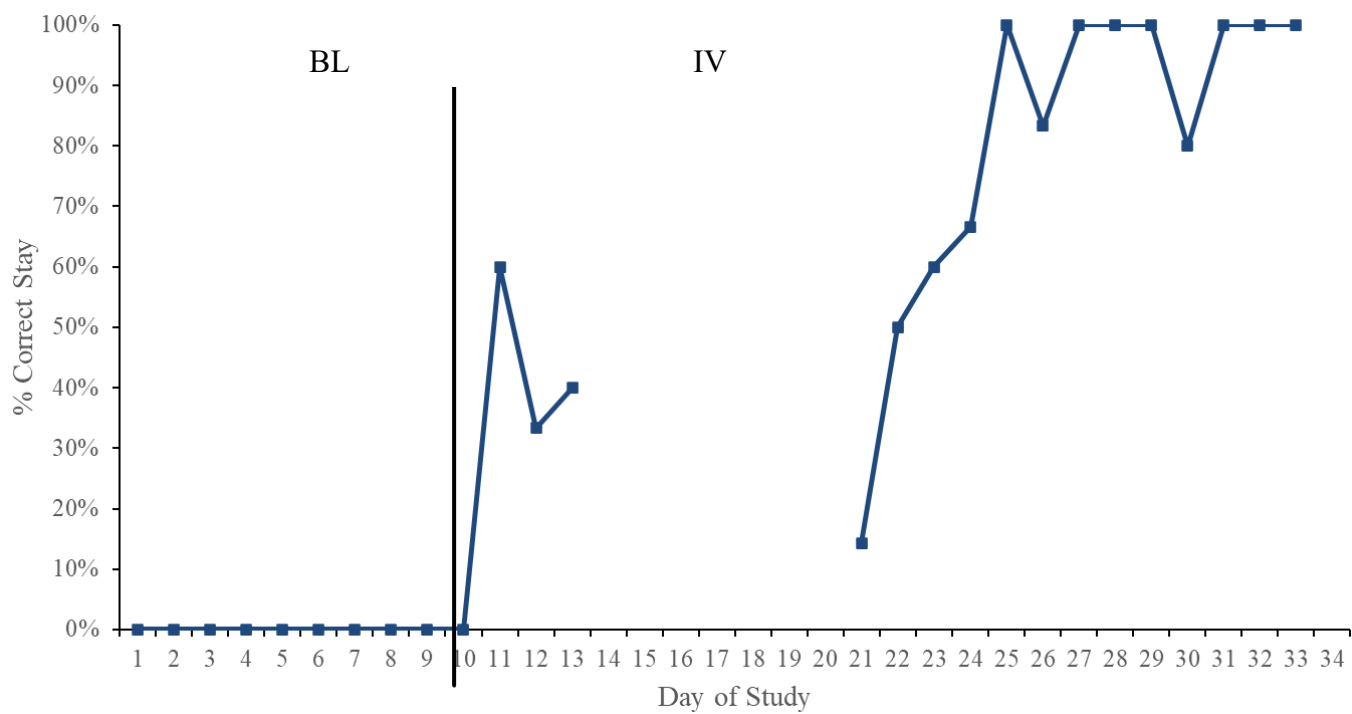


Figure 3

Percentage of Correct Responses for Back-up Behavior Across Baseline (BL) and Intervention (IV)

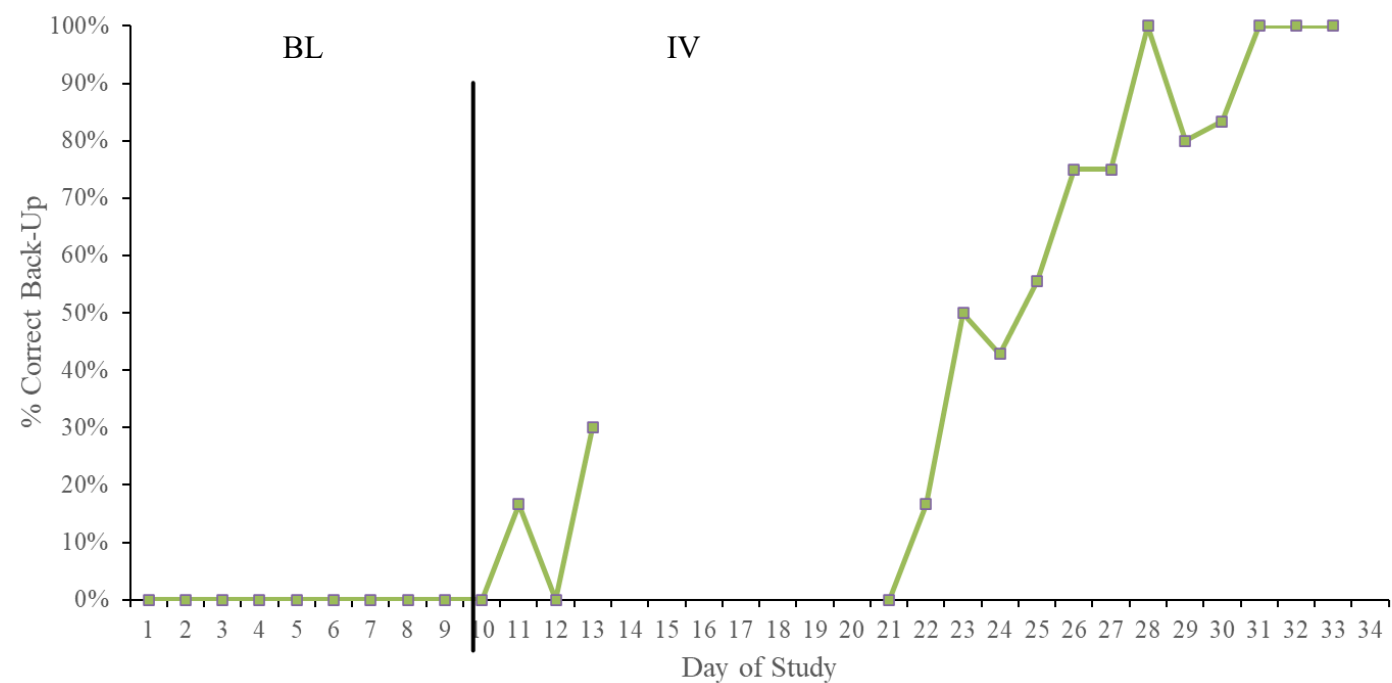


Figure 4

Percentage of Correct Responses for Scratch Behavior Across Baseline (BL) and Intervention (IV)

